

Takes a Village Proposal

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1 Introduction

The lab has gleaned a lot of social metric insights from the Junior High Game. Now I am beginning to wonder how social behavior works with more than one metric, notably a resource metric. Resources are plentiful, but limited in geographic space. I think that the resource and social metrics are influenced by each other. As resources are traded, those interactions lead to an increase or decrease in social standing, and as social standing is established, opportunities for gaining resources decreases.

I have no prior experience making a human society model from scratch. Since it is something I would like to be able to do, I include my current understanding of the topic so I can be corrected and have gaps in my assumptions pointed out.

2 Requisites of a Human Society Model

In our model, we want to study collective action, where groups form naturally to strengthen the members beyond the strength they can achieve on their own. The game model that can do this includes: "a large and flexible strategy space, conflicting interests, high interdependence, power asymmetry, non-convergence, and scalability" (Skaggs and Richards, 2024). These traits are all a part of the Junior High Game, which models them excellently.

Why a new game is needed comes from the question of how a social metric plays with a resource metric - when the agents leave junior high and have to feed themselves. The combination of social power and material power and how to leverage them together to survive is a key part of human life. Since we are trying to model this human social behavior, a second metric is important because, "social behaviour is an exchange of material and nonmaterial (e.g., symbols of approval and prestige) goods" (Treviño, 2009). All human actions and interactions are these exchanges, and parties form sentiments about the other party from these actions. From the above cited work, "Sentiment is behaviour expressive of a person's attitudes toward other persons and includes the "liking and disliking for individuals, approval and disapproval of the things they do". Each agent in a Game Model should have the ability to interact with other agents in this material and nonmaterial way. This duality of the two

interchangeable resources lets us see how having two metrics, not just one, impacts collective action.

3 Takes a Village Game

Takes a Village is a game where all the agents are spawned in an undeveloped area of raw resources. This area will be their village, and it's borders for now are finite. The game has resource value from the raw resources that the players are able to extract. A player is only able to gather one type of resource in the beginning, so they will have to trade with other players who have specialized in the other resources in order to survive and advance. The players need to do this since not consuming food and wood will lead to death and sickness. As players trade, they will form opinions of the other players. These opinions are called sentiment, and players will get to choose what sentiment they want to set for each other player. This means each player will have their own set of sentiments about the other players, each different from the sentiments that the other players have. This internal social score is how humans work. I think this is important in contrast to having a global social metric that averages everything out because what one person does to another will be seen as criminal to some, and heroic to others (current US politics today). This social metric is explained more below.

As players form trade groups and gather more resources, they are able to build more 'developments', which extract more resources for them per day. We'll begin to see how a resource not needed to live, just to advance (like metal) becomes a very powerful resource later in the game as players are vying to develop the rest of the finite resources.

Players will be able to cheat each other in trades, spread false rumors about each other, and seize the developments of other players. Peace will look like the steady growth of all the players, war will look like slander and groups fighting over control of certain developments.

3.1 How the Game is Played

A day is split into three rounds, work, trade, and rumor (Table 1). These phases are played through in order each day, for some number of days. As the players play the phases, they are motivated to do what they need to in order to stay alive. At the end of each day, the players consume one food and burn one wood. For every day that a player does not eat, their daily eating percentage goes down. Once their percentage drops below 0.5 they starve to death. Players start the game with enough food for a few days. For every day that they do not burn wood, a percentage of getting sick increases. Getting sick will leave you unable to work in the working phase for a number of days. This still leaves you the ability to trade for food, but will weaken your financial standing since you are not getting any new resources. Getting sick can happen randomly to anyone, even if they are burning wood, but at a much smaller chance.

Players will have to specialize, and then find a way to get food and wood every day through trading honestly or dishonestly.

Phase	Details
Work	Players work in their development, or build a development. If they work in a development they get resources added to their inventory. Building a development does not give them any resources. Players can also seize a property. They can also join a group currently seizing a property or countering a seize.
Trade	Player will see a list of other players with the player name, the sentiment the player has towards the other player, and potentially the global sentiment in a later version. During trading, a player can advertise a certain resource they are willing to part with. This can be true or a lie.
Rumor	During the day, anytime a player interacts with a different player the player can set their sentiment about the different player to either <i>honest</i> , <i>chaotic</i> , <i>sinister</i> . The rumor phase is when players can say that player X is sentiment Y. For each rumor that is spoken during this phase, you can accept the rumor and your sentiment of the rumored player will change to reflect the rumor, or you can reject the rumor, which keeps your sentiment of the rumored player, and decreases your opinion of the rumor spreader. Or you can ignore, and nothing happens to your sentiments.

Table 1: The Phases of a Day

3.1.1 Phases

Work Phase The working phase is the start of a game day. In this phase physical resource is produced and added to a players inventory, or a player develops raw resource by building a development which he can begin benefiting from the next day.

Players get to choose two resources to produce that day. Typically, the two resources will be the same resource from their first development. If I have a farm development, then I can get two food, for example. A player who has multiple developments can choose to take one resource from two different developments.

During the working phase, someone can approach your development and ask to work there. If you agree, then they make two resources of that development type, but they only get one, you getting the other. A development can employ multiple people during a working round. Playing with the numbers here would be fun to look at how men act when they get a little authority and begin exercising unrighteous dominion. If players got three resources, for example, would a development owner require one resource a day or would he demand two since he knows the employee does not have a development. The negotiating for gaining employees and keeping them would be a fun route to see.

Another working phase option is to seize a development or counter a seize. If a player seizes a development, then they keep the owner from getting any resources from that development. Since players are spending the day seizing or countering a development they don't get to gain any resources that day, but they can trade. If a development is seized for long enough it is repossessed by the first seizer (maybe we see pigs take the Animal Farm with

communist promises, or maybe a utopia can be created). Players can choose to join the seizing player and add numbers to the seizer. The owner of the development either has to surrender the development, or engage in a counter effort where he also is not making any resources. The owner won't be able to take back his development until he can get more people on his group opposing a seize than the seizers group. Group members who starve or get sick overnight are dropped from the group count the next day. (We'll have to decide what happens exactly in this situation, how the seizer can be thwarted or supported in their efforts, and the outcomes. I think it is an important mechanic, it allows interesting behaviors to be taken in certain situations, and I think there are situations where people might even side with the person seizing someone else's property given that the owner's reputation is sinister.)

Trade Phase After the work phase there is a trading phase, where players can approach other players to trade for the resources they don't have on hand. Players trade and develop throughout the game trying to avoid death by starvation or sickness. During the trading phase, you are able to "cheat" the trade. A player has the ability after the trade has been agreed upon to change what they will actually give the other person in the trade. You both communicate what you want to trade, and then both parties will "trade". Then the players receive their end of the bargain and see if they have been swindled or not.

Rumor Phase After trading is complete, players enter the last phase of the day which is rumors. A player will see all the rumors that other players are saying about different player's morality. A player won't see the rumors spread about them, at least in this version. As you see all the rumors, you can choose to believe the rumor, which changes the rumored player's morality in your own sentiments. If you choose to disbelieve the person sharing the rumor, then your perception of the person spreading the rumor will go down, or you can dismiss the rumor and no effect will happen. This effect introduces the Balance Theory by Heider. You are able to spread rumors that are true or untrue, and whether you reflect the sentiment of your rumor is up to you.

End of Day After rumors, the day ends. You will be shown the food and wood you are consuming. If people don't have wood and ask to share your fire, you can share with them or not. They don't become more likely to get sick that night if a player shares fire with them.

3.1.2 Mechanics

Chat There will be a way for players to communicate. Since we need to see how AI's play the game, having them chat could be tricky. The chat might look like pre-made messages with blank spaces for the player to change for trade offers or making a promise ('I won't pay you as much but if you work for me I'll share my fire with you,' sorta thing). Playing without an open chat, where players just have interactions and other player's rumors to go off on would a good way to play as well in some cases. I think initially, where we are playing with humans more and solidifying the game, having a chat would be useful.

Setting Sentiment Players you are interacting with will have a little interface where you can choose to set your personal sentiment about them. You can't set the sentiment of players you haven't interacted with personally, unless you choose to believe rumors about them. All personal sentiments begin as chaotic, since you don't know if they are honest, chaotic, or sinister.

3.2 Questions We Can ask with Takes a Village

1. How does being resource rich/poor sway your social standing of others? Are rich people seen on extremes?
2. How does being socially rich/poor impact your ability to gain wealth? If you are seen as a good person, can you convert that into wealth? If you are seen as sinister, who is willing to interact and trade with you?
3. How does being poor and more vulnerable to sickness (in *Bridges Out Of Poverty* they mention daily family crises and drama also take away from impoverished people's ability to focus on income.) impact your efforts to pull yourself up out of poverty?
4. Can the poor rise up and take down the rich? How do they fair after doing this? (oink oink)
5. Are resource rich people more tempted to forsake their social standing with poor players to swindle them to advance than resource poor/middle class? I can see this happening from JHG games I've played.
6. In a world of finite resources/room to expand, can people still rise up together in plenty? If so, why is the tendency to not?
7. How long can someone effectively lie about their wealth? Does lying about your standing give you benefits to growth, or is a modest honest approach better?
8. Do the impoverished help each other, feeling entitled to ask for and take the rewards of someone else in poverty?
9. How do genetic AI's answer these questions?

As things develop, I want to see if people start valuing that social metric, or if the important thing is to just work hard and try to get enough resources to stay alive. We could see people monopolize and develop lots of resources, keeping a working class barely alive underneath them. We might see poor people rise up and challenge rich swindlers and see if the rich care about that at all. Will the metal workers get to decide who is worthy to buy their metal and advance in society, or will they be forced to get food and wood from whoever is well off enough to have extra to spare? I think there is an interesting society that this game can embody.

3.3 Takes a Village Fulfills Requirements

1. Large and Flexible Strategy Space. Players have the ability to choose who to interact with, what resources to develop, and form groups where agents have their own duties to the group.
2. Conflicting Interests. Players are not playing a max-sum game, trying to survive and solidify their stability through wealth.
3. High interdependence. Agents need multiple types of resources to survive, and in the beginning they can only get one resource at a time.
4. Power Asymmetry. The wealthy can afford to do things like pay people to counter seized developments, not get sick or starve, build more developments and propel their wealth. Poor people without a development have to be employed, and have to get by by working a farm and sharing wood.
5. Non convergence. Resource production continues to grow, where the population does not.
6. Scalability. Lots of players can join in the beginning, the village and raw resources scaling to match.
7. Social and Wealth currencies/metrics. Tradable resources, and tradable sentiments.

4 The First Steps and Question

To start, this game can be played on paper in a few small groups of humans for proof of concept. A base game can be planned and built, with some hand-coded simple AI's. Once AI's have generators and math planned out, we can start training some genetic algorithms like the JHG. The seizing mechanic could potentially not be implemented at first if we are worried about complexity.

4.1 First Question

Does Takes a Village provide a model to study how agents fall into and rise out of poverty?

References

- [1] Skaggs, Johnathan; Richards, Michael (2024) '*Fostering Collective Action in Complex Societies Using Community-Based Agents*', IJCAI24
- [2] Treviño, A. Javier (2009) '*George C. Homans, the human group and elementary social behaviour*', The encyclopedia of pedagogy and informal education.